Early diagnosis of extrahepatic biliary atresia in an open-access medical system.

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Biliary atresia (BA) is the most common cause of cholestatic jaundice in infancy. Early diagnosis and surgical management, ideally before 60 days of age, result in improved outcomes. We aimed to determine the age at diagnosis of BA in the Military Health System (MHS) and to compare the age at diagnosis by access to care models. We hypothesized that children with BA receiving primary care in military facilities have an earlier age at diagnosis due to decreased economic and access barriers. Data for all Tricare enrollees born in fiscal years 2004-2008 with a diagnosis of BA were extracted from MHS databases. Non-parametric tests, Kaplan-Meier curves and log rank tests compared differences in age at diagnosis by type of primary care facility, gender, prematurity and presence of additional anomalies. 64 subjects were identified within the five year period. Median age at diagnosis was 40 days [range 1-189], with 67% diagnosed by 60 days and 80% by 90 days. 45 (70%) received civilian primary care within the MHS. There was no difference in the median age at diagnosis between subjects in the MHS with civilian primary care vs. military primary care (37 days [1-188] vs. 46 days [1-189]; p=0.58). In the MHS, two-thirds of infants with biliary atresia are diagnosed prior to 60 days of life. Gender, prematurity or presence of additional anomalies do not affect the timing of diagnosis. Civilian and military primary care models within the MHS make timely diagnoses of biliary atresia at equivalent rates.

Comparative Demography of Biliary Atresia and Choledochal Malformation in London.

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Choledochal malformation (CM) and biliary atresia (BA) are the two most important bile duct pathologies arising in infancy and childhood. The aim was to investigate for evidence of shared demographic features in a common temporo-spatial area. Patients identified prospectively and defined as being born within metropolitan London in the period 1999-2022. Index of Multiple Deprivation (IMD) Rank Index (based on maternal post-code - England 2019) and live-birth data taken from the Office of National Statistics to assess incidence and differences assessed using the Poisson distribution and Chi2 tests. $P \leq 0.05$ was regarded as significant. 280 patients (BA n = 202; CM n = 78). The incidence of BA and CM in London was 7.45/100,000 (1 in 13,430) and 2.9/100,000 (i.e. 1 in 34,480) with no change over the period. The incidence of BA in Inner London was significantly higher at 9.42/100,000 (1 in 10,650) ($P = 0.006$). Marked female predominance evident in the CM group (77 % vs. 54.4 %; $P = 0.0006$) with more patients of non-white ethnicity in both groups (BA = 52.4 %; CM = 55 %, $P = 0.32$). No difference in IMDR between BA and CM (11,557 vs. 11,594; $P = 0.26$), with a marked difference between Inner and Outer boroughs for CM (8976 vs. 16,717; $P < 0.001$), but less so in BA (10,384 vs. 12,891; $P = 0.039$). IMDR was higher in those of white ethnicity for BA (11,979 vs. 10,660; $P = 0.03$). Inner London has the highest incidence of BA than anywhere in Europe or North America. The calculated incidence of CM is higher than anticipated and the first to be based on actual observation.

[Bile duct atresia: outline for a solution].

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Biliary atresia continues to be a serious and relatively rare disease (1/50,000 newborns) and whose long-term prognosis has changed drastically since the appearance of liver transplant (LT) as a therapeutic weapon. The combination of two factors, early diagnosis and correct application of Kasai's surgical technique, is essential to obtain acceptable results and sufficient biliary drainage allowing the children to overcome the critical 7 kg barrier and place them in the lesser morbi-mortality range in relation to a possible LT. But we must keep in mind that despite its critics, Kasai's technique can guarantee, both in our own experience and in the literature, ten years survival percentages over 50% with correct hepatic function, as well as clinical normality and a quality of life clearly superior to first years post-LT. We present the evolution of a group of 20 patients affected with biliary atresia, diagnosed in our center since 1985, the year when pediatric LT began to be used as a therapeutic procedure in this country. We valued the age of intervention, technique, immediate and long-term results and the evolution and necessity of LT. All 20 patients were analyzed individually, and they currently have an age range from 2-14 years and were all operated by Kasai's technique. We classified the patients as having good, regular or poor results with regards to biliary flow, normalization of billirubin levels and clinical evolution. Sixteen patients presented biliary flow of such an extent that 14 of them, classified as good, completely normalized the billirubin levels. Two others, presently aged 14 and 8 years respectively, present average levels of 2.5-5.5 mg/100 ml and are classified as regular in a situation of advisable transplant, although with an acceptable hepatic function. Only one case, the first in the poor group, did not initially present biliary elimination and died at age six months while on the waiting list. Three other cases in the same group presented insufficient biliary elimination and were transplanted with 7, 11 and 12.5 kg, respectively. The second died in the first year post-transplant. In our opinion, action in biliary atresia must be early and based on the correct application of Kasai's technique, seeking to achieve a biliar flow that eliminates or distances the patient as far as possible from the necessity of a future LT. Three lines come together to obtain this target: an early diagnosis, a correct application of Kasai's technique, and an implication in the follow-up and treatment of these children by the hepatic transplant groups. All this advises us, as is done in other countries, to create reference centers for the study of neonatal cholestasis where an accumulated experience of a relatively rare pathology can be taken advantage of.

Assessment of the nutritional status of infants and children with biliary atresia.

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Since liver transplantation for children with end-stage liver disease has become accepted therapy clinicians have shown interest in the nutritional depletion that occurs with biliary atresia, since children with this condition form the largest group presenting for possible transplantation. Eleven infants and children with biliary atresia (age range 1,5 months-7 years) seen at the Red Cross War Memorial Children's Hospital over a 4-month period (March-June 1988) were studied. All but one had severe cholestasis at the time of the study. Nutrient intake during the hospitalised period was noted. Clinical nutritional parameters were documented and serum levels of vitamins A, D, E and the trace elements zinc and copper were measured. All patients with cholestasis (10/11) over 3 months of age (8/10) showed evidence of severe growth stunting, with weights below the 3rd percentile. Head circumference measurements were less than 5th percentile in 7/8 of those over 4 months of age including a 7-year-old child who had lost his jaundice after porto-enterostomy at age 2 months. All those with cholestasis showed evidence of fat-soluble vitamin deficiency and to a lesser extent zinc deficiency, but had raised serum copper levels. Three of 4 patients receiving cholestyramine had very low levels of vitamin E despite supplementation. These findings confirm the presence of severe nutritional depletion and growth stunting in patients with biliary atresia and the failure of 'normal' nutrient and vitamin supplementation to correct these deficiencies. The importance of close attention to nutrition in adequately preparing those patients assessed as suitable for liver transplantation therapy is stressed.

Biliary Atresia/Neonatal Cholestasis: What is in the Horizon?

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"Biliary atresia (BA) is a common cause of jaundice in infancy. There is increasing evidence that newborn screening with direct or conjugated bilirubin leads to earlier diagnosis. Although the Kasai portoenterostomy is the primary treatment, there are scientific advances in adjuvant therapies. As pediatric patients transition to adult care, multidisciplinary care is essential, given the complexity of this patient population."

Age at surgery and native liver survival in biliary atresia: a systematic review and meta-analysis.

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Biliary atresia (BA) is a childhood rare disease of the liver and bile ducts that requires prompt surgical intervention. Age at surgery is an important prognostic factor; however, controversy exists with regard to the benefit of early Kasai procedure (KP). We aimed to conduct a systematic review and meta-analysis to examine the relationship between the age at KP and native liver survival (NLS) of BA patients. We performed the electronic database search using Pubmed, EMBASE, Cochrane, and Ichushi Web and included all relevant studies published from 1968 up to May 3, 2022. Studies that examined the timing of KP at ages 30, 45, 60, 75, 90, 120, and/or 150 days were included. The outcome measures of interest were NLS rates at 5, 10, 15, 20, and 30 years post-KP and the hazard ratio or risk ratio for NLS. The quality assessment was used using the ROBINS-I tool. Among 1653 potentially eligible studies, nine articles met the inclusion criteria for the meta-analysis. Meta-analysis for hazard ratios revealed that there was a significantly faster time to liver transplantation in the group of patients who had KP at later timing as compared with earlier KP (HR = 2.12, 95% CI 1.51-2.97). The risk ratio comparing KP ≤ 30 days and KP ≥ 31 days on native liver survival was 1.22 (95% CI 1.13-1.31). The sensitivity analysis showed that comparing KP ≤ 30 days and KP 31-60 days, the risk ratio was 1.13, 95% CI 1.04-1.22. Conclusion: Our meta-analysis showed the importance of early diagnosis and surgical interventions ideally before 30 days of life in infants with BA on native liver survival on 5, 10, and 20 years. Therefore, effective newborn screening of BA targeting KP ≤ 30 days is needed to ensure prompt diagnosis of affected infants. What is Known: • Age at surgery is an important prognostic factor. What is New: • Our study performed an updated systematic review and meta-analysis to examine the relationship between age at Kasai procedure and native liver survival in patients with BA.

Biliary atresia: the timing needs a changin'.

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Biliary atresia (BA), a uniquely pediatric liver disease, is the leading cause of liver-related death in children and the most frequent indication for liver transplantation in the pediatric population. Early intervention with a Kasai procedure (KP) is the current standard of care for this condition. The single most important and well-established prognostic factor for the KP outcome is the patient's age at the time of the KP. The older the infant, the less successful the operation and the less favourable is the post-KP survival with native liver. There remains in Canada, and throughout the world, a problem of late referral, delayed diagnosis and older age at surgery. Early disease detection and intervention has been hampered by the lack of an effective screening strategy for BA. Recently, however, novel programs for the early identification of BA in the first month of life, but after two weeks of age, have been successfully implemented and evaluated in some countries, with significantly improved outcomes for affected infants. Whether any of these programs should be adopted to improve the timing of referral and treatment for Canadian infants affected with this devastating liver disease deserves consideration and study.

Biliary atresia.

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Biliary atresia (BA) is a congenital obliterative cholangiopathy of unknown aetiology, affecting both the intra- and extrahepatic bile ducts. Although relatively rare, BA must be excluded in any infant with conjugated hyperbilirubinaemia since the prognosis is improved by early diagnosis and prompt surgery. At least two phenotypes of BA are currently recognized; the syndromic variety is associated with other congenital anomalies and a poorer outcome. The results of treatment have steadily improved and, with a combination of timely expert surgery (Kasai portoenterostomy) and liver transplantation in specialist centres, good quality long-term survival is now possible in more than 90% of affected patients. A better understanding of the aetiology of BA and the pathogenesis of hepatic fibrosis is needed in order to develop new therapeutic strategies.

Biliary atresia with associated structural malformations in Canadian infants.

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Biliary atresia (BA) is associated with extrahepatic congenital malformations in a minority of affected infants. The term commonly applied to this subgroup is 'BASM' for biliary atresia splenic malformation syndrome, as spleen abnormalities are prominent. To examine clinical outcome in Canadian BA patients with extrahepatic congenital malformations in the Canada-wide BA database of patients born between 1985 and 2002, and additionally, to recharacterized the syndrome. Patients had ≥1 of the following: a/polysplenia, abnormal abdominal situs, intestinal malrotation, abdominal vascular anomaly or congenital heart disease. Among 328 BA patients, 44 (13%) had associated congenital abnormalities. Intra-abdominal anomalies included polysplenia (n=25), abnormal abdominal situs (n=9), intestinal malrotation (n=19), portal vein anomaly (n=12), hepatic artery anomaly (n=3) and inferior vena cava interruption (n=20). Twenty-six patients had cardiac malformations including pulmonary stenosis (n=11), ventricular septal defect (n=10), atrial septal defect (n=7), total anomalous pulmonary venous return (n=3), double outlet right ventricle (n=3), tetralogy of Fallot (n=2), atrioventricular canal (n=2), dextrocardia (n=2), bicuspid aortic valve (n=2), hypoplastic left heart (n=1) and partial anomalous pulmonary venous return (n=1). Age at Kasai operation, performance of liver transplant, overall survival, post-Kasai native liver survival and transplant survival were comparable to isolated BA. Presence of polysplenia or complex cardiac disease did not reduce post-Kasai native liver survival. Three patients had

≥2 typical abnormalities without polysplenia: thus, splenic malformations are not essential to this BA subgroup. Hierarchical cluster analysis demonstrated characteristic abnormalities grouped in a multiplicity of combinations, consistent with a spectrum of defective lateralization. We suggest that the acronym 'BASM' be redefined as 'biliary atresia structural malformation'.

Neonatal cholestasis syndrome: Indian scene.

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Neonatal cholestasis syndrome (NCS) in India has largely remained ignored. Three questions need to be addressed: (a) What is known about NCS in India (b) Where do we stand and (c) What needs to be done? Current data on etiology of NCS indicates that biliary atresia contributes to about 40% of all NCS cases. There is considerable delay in the referral of patients to appropriate centres for management. A delay of 120.8 +/- 60.5 days in biliary atresia and 65.9 +/- 39.2 days in neonatal hepatitis were documented. Biliary atresia cases need to be diagnosed and operated by eight weeks of age so as to have the best results. Delayed referral after 3 months of age, not only bring down the success rate considerably but also adversely affect the management with regard to surgical procedures, nutritional support, control of ascites and finally the cost. Cirrhosis rapidly develops in children with biliary atresia. At this stage the only option left is liver transplantation. An important obstacle in the care of infants with NCS is the misconception of jaundice in newborns. This needs to be handled at a professional level in the training of undergraduates and postgraduates and the lay public. Public awareness campaigns like "Yellow Alert" may be useful in this direction.
